GATE CSE 2024 SET-1

65 questions · 100 marks · 180 minutes · MCQ 29, MSQ 20, NAT 16 · MCQ wrong answers cost 1/3 of their marks; MSQ and NAT have no negative marking.

Q1. MCQ · 1 mark, –0.33 · Discrete Mathematics / Functions

Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that $f(x) = \max\left\{x, x^3\right\}, x \in \mathbb{R}$, where $\mathbb{R}$ is the set of all real numbers. The set of all points where $f(x)$ is NOT differentiable is

(A) $-1, 1, 2$

(B) $-2, -1, 1$

(C) $0, 1$

(D) $-1, 0, 1$

Q2. MCQ · 1 mark, –0.33 · Engineering Mathematics / Linear Algebra

The product of all eigenvalues of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ is

(A) -1

(B) 0

(C) 1

(D) 2

Q3. MCQ · 1 mark, –0.33 · Digital Logic / Number System

Consider a system that uses 5 bits for representing signed integers in 2's complement format. In this system, two integers A and B are represented as $A = 01010$ and $B = 11010$. Which one of the following operations will result in either an arithmetic overflow or an arithmetic underflow?

(A) $A + B$

(B) $A - B$

(C) $B - A$

(D) $2 * B$

Q4. MCQ · 1 mark, –0.33 · Discrete Mathematics / Probability Theory

Consider a permutation sampled uniformly at random from the set of all permutations of $\{1, 2, 3, \cdots, n\}$ for some $n \geq 4$. Let X be the event that 1 occurs before 2 in the permutation, and Y the event that 3 occurs before 4 . Which one of the following statements is TRUE?

(A) The events X and Y are mutually exclusive

(B) The events X and Y are independent

(C) Either event X or Y must occur

(D) Event X is more likely than event Y

Q5. MCQ · 1 mark, –0.33 · Computer Organization and Architecture / IO Interface

Which one of the following statements is FALSE?

(A) In the cycle stealing mode of DMA, one word of data is transferred between an I/O device and main memory in a stolen cycle

(B) For bulk data transfer, the burst mode of DMA has a higher throughput than the cycle stealing mode

(C) Programmed I/O mechanism has a better CPU utilization than the interrupt driven I/O mechanism

(D) The CPU can start executing an interrupt service routine faster with vectored interrupts than with non-vectored interrupts

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Q6. MCQ · 1 mark, –0.33 · Computer Networks / Application Layer Protocols

A user starts browsing a webpage hosted at a remote server. The browser opens a single TCP connection to fetch the entire webpage from the server. The webpage consists of a top-level index page with multiple embedded image objects. Assume that all caches (e.g., DNS cache, browser cache) are all initially empty. The following packets leave the user's computer in some order.

(i) HTTP GET request for the index page
(ii) DNS request to resolve the web server's name to its IP address
(iii) HTTP GET request for an image object
(iv) TCP SYN to open a connection to the web server

Which one of the following is the CORRECT chronological order (earliest in time to latest) of the packets leaving the computer ?

(A) (iv), (ii), (iii), (i)

(B) (ii), (iv), (iii), (i)

(C) (ii), (iv), (i), (iii)

(D) (iv), (ii), (i), (iii)

Q7. MCQ · 1 mark, –0.33 · Algorithms / Asymptotic Notation

Given an integer array of size N, we want to check if the array is sorted (in either ascending or descending order). An algorithm solves this problem by making a single pass through the array and comparing each element of the array only with its adjacent elements. The worst-case time complexity of this algorithm is

(A) both $\mathrm{O}(N)$ and $\Omega(N)$

(B) $\mathrm{O}(N)$ but not $\Omega(N)$

(C) $\Omega(N)$ but not $\mathrm{O}(N)$

(D) neither $\mathrm{O}(N)$ nor $\Omega(N)$

Q8. MCQ · 1 mark, –0.33 · Programming and Data Structures / Arithmetic Operation

Consider the following C program:

```
#include <stdio.h>
int main(){
int a = 6;
int b = 0;
while(a < 10) {
a = a / 12 + 1;
a += b;}
printf("%d", a);
return 0;}
```

Which one of the following statements is CORRECT?

(A) The program prints 9 as output

(B) The program prints 10 as output

(C) The program gets stuck in an infinite loop

(D) The program prints 6 as output

Q9. MCQ · 1 mark, –0.33 · Programming and Data Structures / Function

Consider the following C program:

```
#include <stdio.h>
void fX();
int main(){
fX();
return 0;}

void fX(){
char a;
if((a=getchar()) != '\n')
fX();
if(a != '\n')
putchar(a);}
```

Assume that the input to the program from the command line is 1234 followed by a newline character. Which one of the following statements is CORRECT?

(A) The program will not terminate

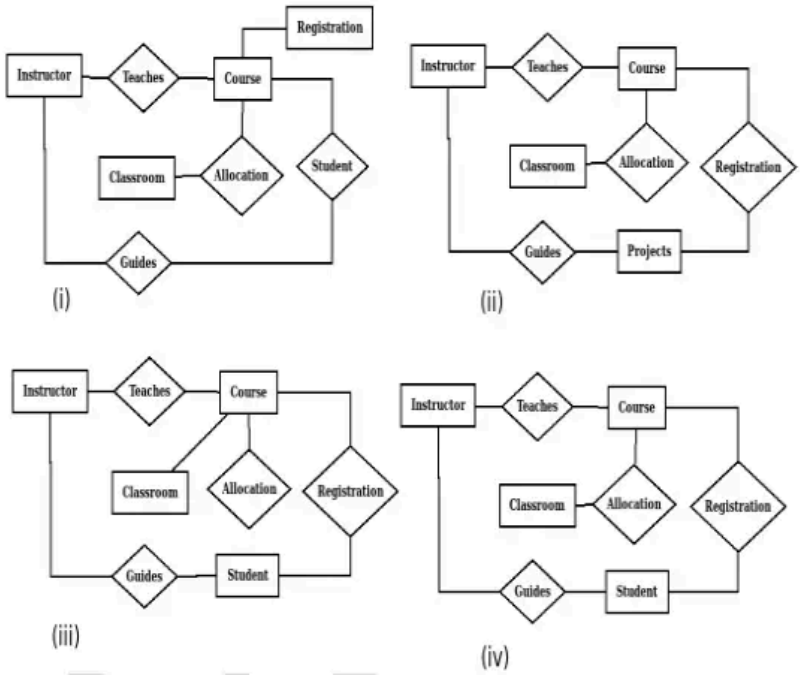
(B) The program will terminate with no outpu

(C) The program will terminate with 4321 as output

(D) The program will terminate with 1234 as output

Q10. MCQ · 1 mark, –0.33 · Databases / ER Model

Let S be the specification: "Instructors teach courses. Students register for courses. Courses are allocated classrooms. Instructors guide students." Which one of the following ER diagrams CORRECTLY represents S?



(A) (i)

(B) (ii)

(C) (iii)

(D) (iv)

Q11. MCQ · 1 mark, –0.33 · Programming and Data Structures / B plus Tree

In a B+ tree, the requirement of at least half-full (50%) node occupancy is relaxed for which one of the following cases?

(A) Only the root node

(B) All leaf nodes

(C) All internal nodes

(D) Only the leftmost leaf node

Q12. MSQ · 1 mark · Databases / Normal Form

Which of the following statements about a relation R in first normal form (1NF) is/are TRUE ?

(A) R can have a multi-attribute key

(B) R cannot have a foreign key

(C) R cannot have a composite attribute

(D) R cannot have more than one candidate key

Q13. MSQ · 1 mark · Theory of Computation / Regular Language

Let L_1, L_2 be two regular languages and L_3 a language which is not regular. Which of the following statements is/are always TRUE?

(A) $L_1 = L_2$ if and only if $L_1 \cap \overline{L_2} = \phi$

(B) $L_1 \cup L_3$ is not regular

(C) $\overline{L_3}$ is not regular

(D) $\overline{L_1} \cup \overline{L_2}$ is regular

Q14. MSQ · 1 mark · Operating System / Thread

Which of the following statements about threads is/are TRUE?

(A) Threads can only be implemented in kernel space

(B) Each thread has its own file descriptor table for open files

(C) All the threads belonging to a process share a common stack

(D) Threads belonging to a process are by default not protected from each other

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Q15. MSQ · 1 mark · Operating System / Process

Which of the following process state transitions is/are NOT possible?

(A) Running to Ready

(B) Waiting to Running

(C) Ready to Waiting

(D) Running to Terminated

Q16. MSQ · 1 mark · Compiler Design / Parsing

Which of the following is/are Bottom-Up Parser(s)?

(A) Shift-reduce Parser

(B) Predictive Parser

(C) LL(1) Parser

(D) LR Parser

Q17. MSQ · 1 mark · Discrete Mathematics / Probability Theory

Let A and B be two events in a probability space with $P(A) = 0.3, P(B) = 0.5$, and $P(A \cap B) = 0.1$. Which of the following statements is/are TRUE?

(A) The two events A and B are independent

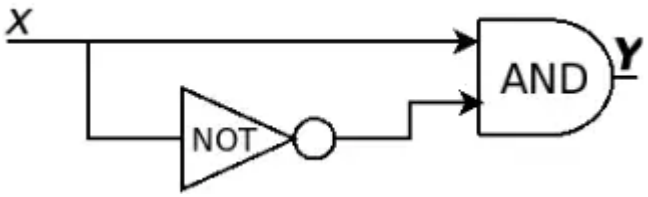
(B) $P(A \cup B) = 0.7$

(C) $P\left(A \cap B^c\right) = 0.2$, where B^c is the complement of the event B

(D) $P\left(A^c \cap B^c\right) = 0.4$, where A^c and B^c are the complements of the events A and B, respectively

Q18. MSQ · 1 mark · Digital Logic / Combinational Circuit

Consider the circuit shown below where the gates may have propagation delays. Assume that all signal transitions occur instantaneously and that wires have no delays. Which of the following statements about the circuit is/are CORRECT?



(A) With no propagation delays, the output Y is always logic Zero

(B) With no propagation delays, the output Y is always logic One

(C) With propagation delays, the output Y can have a transient logic One after X transitions from logic Zero to logic One

(D) With propagation delays, the output Y can have a transient logic Zero after X transitions from logic One to logic Zero

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Q19. MSQ · 1 mark · Computer Networks / Transport Layer Protocol

TCP client P successfully establishes a connection to TCP server Q. Let N_P denote the sequence number in the SYN sent from P to Q. Let N_Q denote the acknowledgement number in the SYN ACK from Q to P. Which of the following statements is/are CORRECT?

(A) The sequence number N_P is chosen randomly by P

(B) The sequence number N_P is always 0 for a new connection

(C) The acknowledgement number N_Q is equal to N_P

(D) The acknowledgement number N_Q is equal to $N_P + 1$

Q20. MSQ · 1 mark · Computer Organization and Architecture / Pipeline Processor

Consider a 5-stage pipelined processor with Instruction Fetch (IF), Instruction Decode (ID), Execute (EX), Memory Access (MEM), and Register Writeback (WB) stages. Which of the following statements about forwarding is/are CORRECT?

(A) In a pipelined execution, forwarding means the result from a source stage of an earlier instruction is passed on to the destination stage of a later instruction

(B) In forwarding, data from the output of the MEM stage can be passed on to the input of the EX stage of the next instruction

(C) Forwarding cannot prevent all pipeline stalls

(D) Forwarding does not require any extra hardware to retrieve the data from the pipeline stages

Q21. MSQ · 1 mark · Computer Networks / Network Layer Protocol

Which of the following fields is/are modified in the IP header of a packet going out of a network address translation (NAT) device from an internal network to an external network?

(A) Source IP

(B) Destination IP

(C) Header Checksum

(D) Total Length

Q22. NAT · 1 mark · Discrete Mathematics / Functions

Let A and B be non-empty finite sets such that there exist one-to-one and onto functions (i) from A to B and (ii) from $A \times A$ to $A \cup B$. The number of possible values of $|A|$ is ___

Answer: _____

Q23. NAT · 1 mark · Programming and Data Structures / Stack

Consider the operator precedence and associativity rules for the integer arithmetic operators given in the table below.

$Operator$	$Precedence$	$Associativity$
$+$	$Highest$	$Left$
$-$	$High$	$Right$
$*$	$Medium$	$Right$
$/$	Low	$Right$

The value of the expression $3 + 1 + 5 * 2/7 + 2 - 4 - 7 - 6/2$ as per the above rules is _____

Answer: _____

Q24. NAT · 1 mark · Algorithms / Minimum Spanning Tree

The number of spanning trees in a complete graph of 4 vertices labelled A, B, C, and D is _____

Answer: _____

Q25. NAT · 1 mark · Databases / Tuple Calculus

Consider the following two relations, $R(A, B)$ and $S(A, C)$:

R

A	B
10	20
20	30
30	40
30	50
50	95

S

A	C
10	90
30	45
40	80

The total number of tuples obtained by evaluating the following expression $\sigma_{B<C}\left(R \bowtie_{R.A=S.A} S\right)$ is ____

Answer: _____

Q26. MCQ · 2 marks, –0.67 · Computer Networks / Data Link Layer

Consider a network path $\mathrm{P}-\mathrm{Q}-\mathrm{R}$ between nodes P and R via router Q. Node P sends a file of size 10^6 bytes to R via this path by splitting the file into chunks of 10^3 bytes each. Node P sends these chunks one after the other without any wait time between the successive chunk transmissions. Assume that the size of extra headers added to these chunks is negligible, and that the chunk size is less than the MTU.

Each of the links $\mathrm{P}-\mathrm{Q}$ and $\mathrm{Q}-\mathrm{R}$ has a bandwidth of $10^6 \mathrm{bits/sec}$, and negligible propagation latency. Router Q immediately transmits every packet it receives from P to R, with negligible processing and queueing delays. Router Q can simultaneously receive on link $P-Q$ and transmit on $\mathrm{link}\, Q-R$.

Assume P starts transmitting the chunks at time $t=0$.
Which one of the following options gives the time (in seconds, rounded off to 3 decimal places) at which R receives all the chunks of the file?

(A) 8

(B) 8.008

(C) 15.992

(D) 16

Q27. MCQ · 2 marks, –0.67 · Compiler Design / Syntax-directed Translation

Consider the following syntax-directed definition (SDD).

$S \rightarrow DHTU$	$\{S.val = D.val + H.val + T.val + U.val;\}$
$D \rightarrow "M" D_1$	$\{D.val = 5 + D_1.val;\}$
$D \rightarrow \epsilon$	$\{D.val = -5;\}$
$H \rightarrow "L" H_1$	$\{H.val = 5 * 10 + H_1.val;\}$
$H \rightarrow \epsilon$	$\{H.val = -10;\}$
$T \rightarrow "C" T_1$	$\{T.val = 5 * 100 + T_1.val;\}$
$T \rightarrow \epsilon$	$\{T.val = -5;\}$
$U \rightarrow "K"$	$\{U.val = 5;\}$

Given "MMLK" as the input, which one of the following options is the CORRECT value computed by the SDD (in the attribute S.val)?

(A) 45

(B) 50

(C) 55

(D) 65

Q28. MCQ · 2 marks, –0.67 · Compiler Design / Parsing

Consider the following grammar G, with S as the start symbol. The grammar G has three incomplete productions denoted by (1), (2), and (3).

$S \rightarrow daT \mid \quad (1)$
$T \rightarrow aS|bT|(2)$
$R \rightarrow \ (3) \ \mid \epsilon$

The set of terminals is $\{a, b, c, d, f\}$. The FIRST and FOLLOW sets of the different non-terminals are as follows.

$$\text{FIRST}(S) = \{c, d, f\}, \text{FIRST}(T) = \{a, b, \epsilon\}, \text{FIRST}(R) = \{c, \epsilon\}$$
$$\text{FOLLOW}(S) = \text{FOLLOW}(T) = \{c, f, \$\}, \text{FOLLOW}(R) = \{f\}$$

Which one of the following options CORRECTLY fills in the incomplete productions?

(A) (1) $S \rightarrow Rf$
(2) $T \rightarrow \epsilon$
(3) $R \rightarrow cTR$

(B) (1) $S \rightarrow fR$
(2) $T \rightarrow \epsilon$
(3) $R \rightarrow cTR$

(C) (1) $S \rightarrow fR$
(2) $T \rightarrow cT$
(3) $R \rightarrow cR$

(D) (1) $S \rightarrow Rf$
(2) $T \rightarrow cT$
(3) $R \rightarrow cR$

Q29. MCQ · 2 marks, –0.67 · Compiler Design / Intermediate Code Generation

Consider the following pseudo-code.

$$
\begin{aligned}
L1: &\quad t1 = -1 \\
L2: &\quad t2 = 0 \\
L3: &\quad t3 = 0 \\
L4: &\quad t4 = 4 * t3 \\
L5: &\quad t5 = 4 * t2 \\
L6: &\quad t6 = t5 * M \\
L7: &\quad t7 = t4 + t6 \\
L8: &\quad t8 = a[t7] \\
L9: &\quad if\ t8 <= max\ goto\ L11 \\
L10: &\quad t1 = t8 \\
L11: &\quad t3 = t3 + 1 \\
L12: &\quad if\ t3 < M\ gotoL4 \\
L13: &\quad t2 = t2 + 1 \\
L14: &\quad if\ t2 < N\ goto\ L3 \\
L15: &\quad max = t1
\end{aligned}
$$

Which of of the following options CORRECTLY specifies the number of basic blocks and the number of instructions in the largest basic block, respectively?

(A) 6 and 6

(B) 6 and 7

(C) 7 and 7

(D) 7 and 6

Q30. MCQ · 2 marks, –0.67 · Operating System / Process Synchronization

Consider the following two threads $\mathrm{T1}$ and $\mathrm{T2}$ that update two shared variables a and b. Assume that initially $\mathrm{a} = \mathrm{b} = 1$. Though context switching between threads can happen at any time, each statement of $\mathrm{T1}$ or $\mathrm{T2}$ is executed atomically without interruption.

T1	T2
$\mathrm{a} = \mathrm{a} + 1;$	$\mathrm{b} = 2 * \mathrm{b};$
$\mathrm{b} = \mathrm{b} + 1;$	$\mathrm{a} = 2 * \mathrm{a};$

Which one of the following options lists all the possible combinations of values of a and b after both $\mathrm{T1}$ and $\mathrm{T2}$ finish execution?

(A) (a=4, b=4) ;(a=3, b=3) ;(a=4, b=3)

(B) (a=3, b=4) ;(a=4, b=3) ;(a=3, b=3)

(C) (a=4, b=4) ;(a=4, b=3) ;(a=3, b=4)

(D) (a=2, b=2) ;(a=2, b=3) ;(a=3, b=4)

Q31. MCQ · 2 marks, –0.67 · Programming and Data Structures / Heap Tree

An array $[82, 101, 90, 11, 111, 75, 33, 131, 44, 93]$ is heapified. Which one of the following options represents the first three elements in the heapified array?

(A) $82, 90, 101$

(B) $82, 11, 93$

(C) $131, 11, 93$

(D) $131, 111, 90$

Q32. MCQ · 2 marks, –0.67 · Algorithms / Recurrence Relation

Consider the following recurrence relation:

$$T(n) = \begin{cases} \sqrt{n}T(\sqrt{n}) + n \text{ for } n \geq 1, \\ \quad 1 \quad \text{for } n = 1. \end{cases}$$

Which one of the following options is CORRECT?

(A) $T(n) = \Theta(n \log \log n)$

(B) $T(n) = \Theta(n \log n)$

(C) $T(n) = \Theta\left(n^2 \log n\right)$

(D) $T(n) = \Theta\left(n^2 \log \log n\right)$

Q33. MCQ · 2 marks, –0.67 · Programming and Data Structures / Heap Tree

Consider a binary min-heap containing 105 distinct elements. Let k be the index (in the underlying array) of the maximum element stored in the heap. The number of possible values of k is

(A) 53

(B) 52

(C) 27

(D) 1

Q34. MSQ · 2 marks · Databases / Relational Schema

The symbol $\rightarrow$ indicates functional dependency in the context of a relational database. Which of the following options is/are TRUE?

(A) $(X, Y) \rightarrow (Z, W)$ implies $X \rightarrow (Z, W)$

(B) $(X, Y) \rightarrow (Z, W)$ implies $(X, Y) \rightarrow Z$

(C) $((X, Y) \rightarrow Z$ and $W \rightarrow Y)$ implies $(X, W) \rightarrow Z$

(D) $(X \rightarrow Y$ and $Y \rightarrow Z)$ implies $X \rightarrow Z$

Q35. MSQ · 2 marks · Discrete Mathematics / Graph Theory

Let G be a directed graph and T a depth first search (DFS) spanning tree in G that is rooted at a vertex v. Suppose T is also a breadth first search (BFS) tree in G, rooted at v. Which of the following statements is/are TRUE for every such graph G and tree T ?

(A) There are no back-edges in G with respect to the tree T

(B) There are no cross-edges in G with respect to the tree T

(C) There are no forward-edges in G with respect to the tree T

(D) The only edges in G are the edges in T

Q36. MSQ · 2 marks · Databases / Transaction

Consider the following read-write schedule S over three transactions T_1, T_2, and T_3, where the subscripts in the schedule indicate transaction IDs:

$$S : r_1(z); w_1(z); r_2(x); r_3(y); w_3(y); r_2(y); w_2(x); w_2(y);$$

Which of the following transaction schedules is/are conflict equivalent to S ?

(A) $T_1T_2T_3$

(B) $T_1T_3T_2$

(C) $T_3T_2T_1$

(D) $T_3T_1T_2$

Q37. MSQ · 2 marks · Digital Logic / Boolean Algebra

Consider a Boolean expression given by $F(X, Y, Z) = \Sigma(3, 5, 6, 7)$.
Which of the following statements is/are CORRECT?

(A) $F(X, Y, Z) = \Pi(0, 1, 2, 4)$

(B) $F(X, Y, Z) = XY + YZ + XZ$

(C) $F(X, Y, Z)$ is independent of input Y

(D) $F(X, Y, Z)$ is independent of input X

Q38. MSQ · 2 marks · Programming and Data Structures / Loop

Consider the following C function definition.

```
int f(int x, int y) {
for (int i=0; i < y; i++) {
x=x+x+y;
}
return x;
}
```

Which of the following statements is/are TRUE about the above function?

(A) If the inputs are $x = 20, y = 10$, then the return value is greater than 2^{20}

(B) If the inputs are $x = 20, y = 20$, then the return value is greater than 2^{20}

(C) If the inputs are $x = 20, y = 10$, then the return value is less than 2^{10}

(D) If the inputs are $x = 10, y = 20$, then the return value is greater than 2^{20}

Q39. MSQ · 2 marks · Engineering Mathematics / Linear Algebra

Let A be any $n \times m$ matrix, where $m > n$. Which of the following statements is/are TRUE about the system of linear equations $Ax = 0$?

(A) There exist at least $m - n$ linearly independent solutions to this system

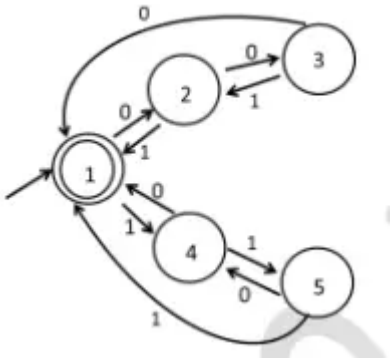
(B) There exist $m - n$ linearly independent vectors such that every solution is a linear combination of these vectors

(C) There exists a non-zero solution in which at least $m - n$ variables are 0

(D) There exists a solution in which at least n variables are non-zero

Q40. MSQ · 2 marks · Theory of Computation / Finite Automata

Consider the 5 -state DFA M accepting the language $L(M) \subset (0+1)^*$ shown below. For any string $w \in (0+1)^*$ let $n_0(w)$ be the number of $0's$ in w and $n_1(w)$ be the number of 1 's in w.



Which of the following statements is/are FALSE?

(A) States 2 and 4 are distinguishable in M

(B) States 3 and 4 are distinguishable in M

(C) States 2 and 5 are distinguishable in M

(D) Any string w with $n_0(w) = n_1(w)$ is in $L(M)$

Q41. MSQ · 2 marks · Discrete Mathematics / Graph Theory

The chromatic number of a graph is the minimum number of colours used in a proper colouring of the graph. Let G be any graph with n vertices and chromatic number k. Which of the following statements is/are always TRUE?

(A) G contains a complete subgraph with k vertices

(B) G contains an independent set of size at least n/k

(C) G contains at least $k(k-1)/2$ edges

(D) G contains a vertex of degree at least k

Q42. MSQ · 2 marks · Discrete Mathematics / Group Theory

Consider the operators $\diamond$ and $\square$ defined by $a \diamond b = a + 2b, a \square b = ab$, for positive integers. Which of the following statements is/are TRUE?

(A) Operator $\diamond$ obeys the associative law

(B) Operator $\square$ obeys the associative law

(C) Operator $\diamond$ over the operator $\square$ obeys the distributive law

(D) Operator $\square$ over the operator $\diamond$ obeys the distributive law

Q43. MSQ · 2 marks · Computer Organization and Architecture / Cache Memory

Consider two set-associative cache memory architectures: WBC, which uses the write back policy, and WTC, which uses the write through policy. Both of them use the LRU (Least Recently Used) block replacement policy. The cache memory is connected to the main memory. Which of the following statements is/are TRUE?

(A) A read miss in WBC never evicts a dirty block

(B) A read miss in WTC never triggers a write back operation of a cache block to main memory

(C) A write hit in WBC can modify the value of the dirty bit of a cache block

(D) A write miss in WTC always writes the victim cache block to main memory before loading the missed block to the cache

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Q44. NAT · 2 marks · Computer Organization and Architecture / Secondary Storage

Consider a 512 GB hard disk with 32 storage surfaces. There are 4096 sectors per track and each sector holds 1024 bytes of data. The number of cylinders in the hard disk is ____

Answer: _____

Q45. NAT · 2 marks · Computer Organization and Architecture / Pipeline Processor

The baseline execution time of a program on a 2 GHz single core machine is 100 nanoseconds (ns). The code corresponding to 90% of the execution time can be fully parallelized. The overhead for using an additional core is 10 ns when running on a multicore system. Assume that all cores in the multicore system run their share of the parallelized code for an equal amount of time.
The number of cores that minimize the execution time of the program is _____

Answer: _____

Q46. NAT · 2 marks · Computer Organization and Architecture / Cache Memory

A given program has 25% load/store instructions. Suppose the ideal CPI (cycles per instruction) without any memory stalls is 2. The program exhibits 2% miss rate on instruction cache and 8% miss rate on data cache. The miss penalty is 100 cycles. The speedup (rounded off to two decimal places) achieved with a perfect cache (i.e., with NO data or instruction cache misses) is _____

Answer: _____

Q47. NAT · 2 marks · Operating System / System Call

Consider the following code snippet using the fork() and wait() system calls.
Assume that the code compiles and runs correctly, and that the system calls run successfully without any errors.

```
int x = 3;
while(x > 0) {
fork();
printf("hello");
wait(NULL);
x--;
}
```

The total number of times the printf statement is executed is ___

Answer: _____

Q48. NAT · 2 marks · Computer Networks / Network Layer Protocol

Consider the entries shown below in the forwarding table of an $\mathbb{P}$ router. Each entry consists of an IP prefix and the corresponding next hop router for packets whose destination IP address matches the prefix. The notation "$/\mathrm{N}$" in a prefix indicates a subnet mask with the most significant N bits set to 1.

$Prefix$	$Next\ hop\ router$
$10.1.1.0/24$	$R1$
$10.1.1.128/25$	$R2$
$10.1.1.64/26$	$R3$
$10.1.1.192/26$	$R4$

This router forwards 20 packets each to 5 hosts. The IP addresses of the hosts are 10.1.1.16, 10.1.1.72, 10.1.1.132, 10.1.1.191, and 10.1.1.205. The number of packets forwarded via the next hop router $R2$ is

Answer: _____

Q49. NAT · 2 marks · Theory of Computation / Context Free Grammar

Let $G = (V, \Sigma, S, P)$ be a context-free grammar in Chomsky Normal Form with $\Sigma = \{a, b, c\}$ and V containing 10 variable symbols including the start symbol S. The string $w = a^{30}b^{30}c^{30}$ is derivable from S. The number of steps (application of rules) in the derivation $S \to^* w$ is _____

Answer: _____

Q50. NAT · 2 marks · Discrete Mathematics / Graph Theory

The number of edges present in the forest generated by the DFS traversal of an undirected graph G with 100 vertices is 40 . The number of connected components in G is ____

Answer: _____

Q51. NAT · 2 marks · Theory of Computation / Regular Expression

Consider the following two regular expressions over the alphabet $\{0, 1\}$:

$r = 0^* + 1^*$

$s = 01^* + 10^*$

The total number of strings of length less than or equal to 5 , which are neither in r nor in s, is ____

Answer: _____

Q52. NAT · 2 marks · Operating System / Memory Management

Consider a memory management system that uses a page size of $2\ \mathrm{KB}$. Assume that both the physical and virtual addresses start from 0. Assume that the pages $0, 1, 2$, and 3 are stored in the page frames $1, 3, 2$, and 0, respectively. The physical address (in decimal format) corresponding to the virtual address 2500 (in decimal format) is ____

Answer: _____

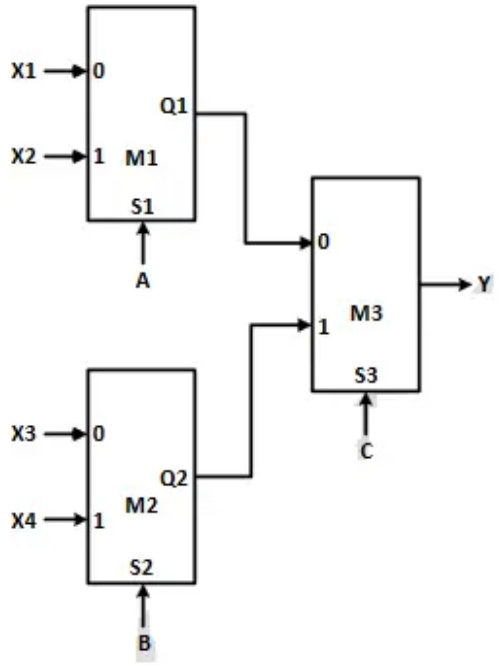
Q53. NAT · 2 marks · Discrete Mathematics / Probability Theory

A bag contains 10 red balls and 15 blue balls. Two balls are drawn randomly without replacement. Given that the first ball drawn is red, the probability (rounded off to 3 decimal places) that both balls drawn are red is ____

Answer: _____

Q54. NAT · 2 marks · Digital Logic / Combinational Circuit

Consider a digital logic circuit consisting of three 2-to-1 multiplexers M1, M2, and M3 as shown below. $X1$ and $X2$ are inputs of $M1$. $X3$ and $X4$ are inputs of $M2$. A, B, and C are select lines of $M1, M2$, and $M3$, respectively.



For an instance of inputs $X\mathbf{Xl} = \mathbf{X2} = \mathbf{1}, \mathrm{X3} = \mathbf{0}$, and $\mathrm{X4} = \mathbf{0}$, the number of combinations of A, B, C that give the output $Y = 1$ is _____

Answer: _____

Q55. NAT · 2 marks · Computer Networks / Network Layer Protocol

Consider sending an IP datagram of size 1420 bytes (including 20 bytes of IP header) from a sender to a receiver over a path of two links with a router between them. The first link (sender to router) has an MTU (Maximum Transmission Unit) size of 542 bytes, while the second link (router to receiver) has an MTU size of 360 bytes. The number of fragments that would be delivered at the receiver is _____

Answer: _____

Q56. MCQ · 1 mark, –0.33 · General Aptitude / Verbal Ability

If '$\rightarrow$' denotes increasing order of intensity, then the meaning of the words
[dry $\rightarrow$ arid $\rightarrow$ parched] is analogous to [diet $\rightarrow$ fast $\rightarrow$ _____].

Which one of the given options is appropriate to fill the blank?

(A) starve

(B) reject

(C) feast

(D) deny

Q57. MCQ · 1 mark, –0.33 · General Aptitude / Numerical Ability

If two distinct non-zero real variables x and y are such that $(x+y)$ is proportional to $(x-y)$ then the value of $\frac{x}{y}$

(A) depends on xy

(B) depends only on x and not on y

(C) depends only on y and not on x

(D) is a constant

Q58. MCQ · 1 mark, –0.33 · General Aptitude / Numerical Ability

Consider the following sample of numbers:

9, 18, 11, 14, 15, 17, 10, 69, 11, 13

The median of the sample is

(A) 13.5

(B) 14

(C) 11

(D) 18.7

Q59. MCQ · 1 mark, –0.33 · General Aptitude / Numerical Ability

The number of coins of Rs. 1, Rs. 5, and Rs. 10 denominations that a person has are in the ratio 5:3:13. Of the total amount, the percentage of money in Rs. 5 coins is

(A) 21%

(B) $14\frac{2}{7}\%$

(C) 10%

(D) 30%

Q60. MCQ · 1 mark, –0.33 · General Aptitude / Numerical Ability

For positive non-zero real variables p and q, if

$$\log\left(p^2+q^2\right) = \log p + \log q + 2\log 3,$$

then, the value of $\frac{p^4+q^4}{p^2q^2}$ is

(A) 79

(B) 81

(C) 9

(D) 83

Q61. MCQ · 2 marks, –0.67 · General Aptitude / Verbal Ability

In the given text, the blanks are numbered (i)-(iv). Select the best match for all the blanks.
Steve was advised to keep his head ___(i)___ before heading __(ii)__ to bat; for, while he had a head __(iii)__ batting, he could only do so with a cool head s __(iv)__ his shoulders

(A) (i) down (ii) down (iii) on (iv) for

(B) (i) on (ii) down (iii) for (iv) on

(C) (i) down (ii) out (iii) for (iv) on

(D) (i) on (ii) out (iii) on (iv) for

Q62. MCQ · 2 marks, –0.67 · General Aptitude / Numerical Ability

A rectangular paper sheet of dimensions 54cm x 4cm is taken. The two longer edges of the sheet are joined together to create a cylindrical tube. A cube whose surface area is equal to the area of the sheet is also taken.
Then, the ratio of the volume of the cylindrical tube to the volume of the cube is

(A) $1/\pi$

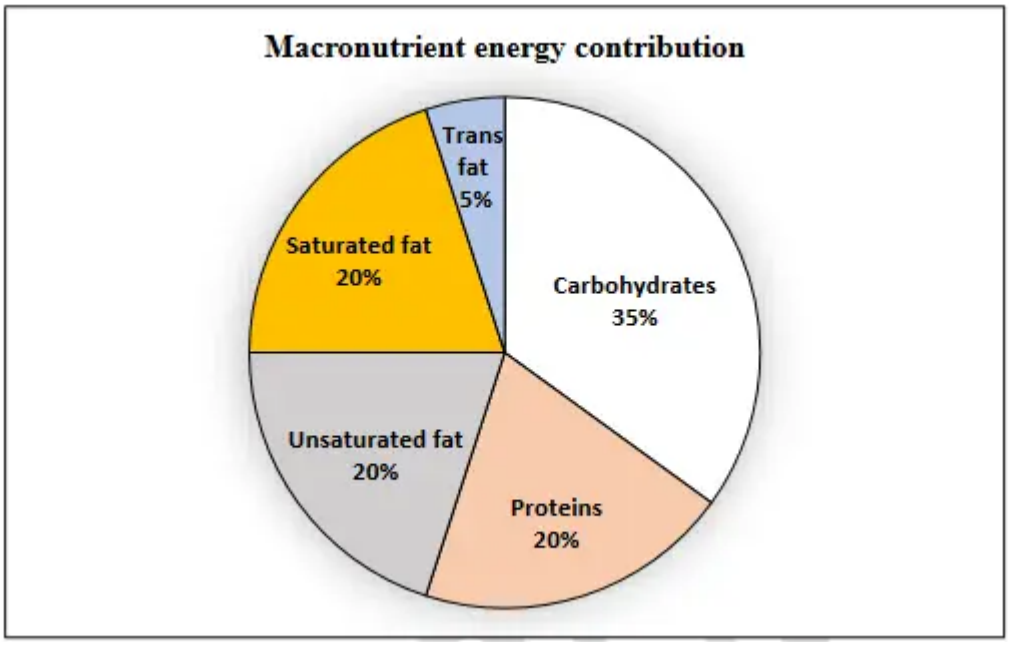
(B) $2/\pi$

(C) $3/\pi$

(D) $4/\pi$

Q63. MCQ · 2 marks, –0.67 · General Aptitude / Numerical Ability

The pie chart presents the percentage contribution of different macronutrients to a typical 2,000 kcal diet of a person.



The typical energy density (kcal/g) of these macronutrients is given in the table.

$Macronutrient$	$Energydensity(kcal/g)$
$Carbohydrates$	4
$Proteins$	4
$Unsaturatedfat$	9
$Saturatedfat$	9
$Transfat$	9

The total fat (all three types), in grams, this person consumes is

(A) 44.4

(B) 77.8

(C) 100

(D) 3600

Q64. MCQ · 2 marks, –0.67 · General Aptitude / Numerical Ability

A rectangular paper of 20cm x 8cm is folded 3 times. Each fold is made along the line of symmetry, which is perpendicular to its long edge. The perimeter of the final folded sheet (in cm) is

(A) 18

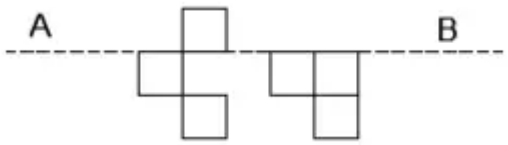
(B) 24

(C) 20

(D) 21

Q65. MCQ · 2 marks, –0.67 · General Aptitude / Numerical Ability

The least number of squares to be added in the figure to make AB a line of symmetry is



(A) 6

(B) 4

(C) 5

(D) 7

Answer key

Q	Type	Answer
Q1	MCQ	D
Q2	MCQ	B
Q3	MCQ	B
Q4	MCQ	B
Q5	MCQ	C
Q6	MCQ	C
Q7	MCQ	A
Q8	MCQ	C
Q9	MCQ	C
Q10	MCQ	D
Q11	MCQ	A
Q12	MSQ	A, C
Q13	MSQ	C, D
Q14	MSQ	D
Q15	MSQ	B, C
Q16	MSQ	A, D
Q17	MSQ	B, C
Q18	MSQ	A, C
Q19	MSQ	A, D
Q20	MSQ	A, B, C
Q21	MSQ	A, C
Q22	NAT	2
Q23	NAT	6
Q24	NAT	16
Q25	NAT	2
Q26	MCQ	B
Q27	MCQ	A
Q28	MCQ	A
Q29	MCQ	D
Q30	MCQ	A
Q31	MCQ	D
Q32	MCQ	A
Q33	MCQ	A
Q34	MSQ	B, C, D
Q35	MSQ	C

Q36	MSQ	B, C, D
Q37	MSQ	A, B
Q38	MSQ	B, D
Q39	MSQ	A
Q40	MSQ	B, C
Q41	MSQ	B, C
Q42	MSQ	B, D
Q43	MSQ	B, C
Q44	NAT	4096
Q45	NAT	3
Q46	NAT	3
Q47	NAT	14
Q48	NAT	40
Q49	NAT	179
Q50	NAT	60
Q51	NAT	44
Q52	NAT	6596
Q53	NAT	0.37 to 0.38
Q54	NAT	4
Q55	NAT	6
Q56	MCQ	A
Q57	MCQ	D
Q58	MCQ	A
Q59	MCQ	C
Q60	MCQ	A
Q61	MCQ	C
Q62	MCQ	A
Q63	MCQ	C
Q64	MCQ	A
Q65	MCQ	A

Solutions

Q1. Answer: D

Compare x and x^3: they are equal at $x = -1, 0, 1$. So
$f(x) = x$ for $x \leq -1$, x^3 on $[-1, 0]$, x on $[0, 1]$, x^3 for $x \geq 1$.

At every switching point the two pieces have different slopes ($\frac{d}{dx}x = 1$, $\frac{d}{dx}x^3 = 3x^2$):
- $x = -1$: slopes 1 and $3 \rightarrow$ corner
- $x = 0$: slopes 0 and $1 \rightarrow$ corner
- $x = 1$: slopes 1 and $3 \rightarrow$ corner

Everywhere else f is a polynomial, hence differentiable.

Answer: $\{-1, 0, 1\}$

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Q2. Answer: B

Product of all eigenvalues $= \det(A)$.

$$\det(A) = 1(45 - 48) - 2(36 - 42) + 3(32 - 35) = -3 + 12 - 9 = 0$$

(Quick check: $R_1 + R_3 = 2R_2$, so the rows are dependent and the matrix is singular.)

Answer: 0

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Q3. Answer: B

With 5-bit 2's complement the range is -16 to $+15$.

$A = 01010 = +10$, $B = 11010 = -6$.

- $A + B = 4 \rightarrow$ fits
- $A - B = 10 + 6 = 16 \rightarrow$ **exceeds +15, overflow**
- $B - A = -16 \rightarrow$ fits (lowest value)
- $2B = -12 \rightarrow$ fits

Answer: A − B

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Q4. Answer: B

By symmetry, 1 is before 2 in half of all permutations, so $P(X) = \frac{1}{2}$; similarly $P(Y) = \frac{1}{2}$.

The relative order of {1,2} and the relative order of {3,4} involve different elements, and all 4 combinations (12/21 with 34/43) are equally likely, so $P(X \cap Y) = \frac{1}{4} = P(X)P(Y) \rightarrow$ **independent**.

- Not mutually exclusive: 1 2 3 4 . . . satisfies both.
- Neither must occur: 2 1 4 3 . . . satisfies none.
- Equally likely, so X is not more likely than Y.

Answer: X and Y are independent

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Q5. Answer: C

Checking each statement:
- Cycle stealing: DMA takes one bus cycle and moves one word — TRUE.
- Burst mode keeps the bus for the whole block (no repeated bus requests), so it has higher throughput for bulk transfer — TRUE.
- **Programmed I/O has better CPU utilization than interrupt-driven I/O** — **FALSE.** In programmed I/O the CPU keeps polling the device status and wastes cycles; interrupt-driven I/O lets the CPU do other work until the device interrupts.
- Vectored interrupts give the ISR address directly (no polling to find the device), so ISR starts faster — TRUE.

Answer: Programmed I/O has better CPU utilization than interrupt driven I/O (FALSE)

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Q6. Answer: C

Order in which the packets leave the computer:
1. **(ii) DNS request** — the IP address of the server is needed before anything else (caches are empty).
2. **(iv) TCP SYN** — open the TCP connection with the server.
3. **(i) GET index page** — first request on the connection.
4. **(iii) GET image** — the browser knows about the image only after it receives and parses the index page.

Answer: (ii), (iv), (i), (iii)

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Q7. Answer: A

The algorithm compares each pair of adjacent elements once: $N - 1$ comparisons in the worst case (when the array is actually sorted, it must check every pair before saying YES).

It cannot do less: any element not examined could break the order. So the worst case is $\Theta(N)$, i.e. both $O(N)$ and $\Omega(N)$.

Answer: both $O(N)$ and $\Omega(N)$

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Q8. Answer: C

Integer division is used.

Iteration 1: $a = 6/12 + 1 = 0 + 1 = 1$, then $a = a + b = 1 + 0 = 1$.
Iteration 2: $a = 1/12 + 1 = 0 + 1 = 1$ again, and so on.

a is stuck at 1, so the condition $a < 10$ is always true and printf is never reached.

Answer: The program gets stuck in an infinite loop

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Q9. Answer: C

Each call of fX() reads one character and, if it is not newline, calls fX() again *before* printing. Each call has its own local a.

Calls read: '1' $\rightarrow$ '2' $\rightarrow$ '3' $\rightarrow$ '4' $\rightarrow$ '\n'. The last call reads newline, makes no further call and prints nothing.

While returning, the calls print their characters in reverse order: 4, 3, 2, 1. Then main returns normally.

Answer: The program will terminate with 4321 as output

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Q10. Answer: D

From the specification we need 4 entity sets — Instructor, Course, Student, Classroom — and 4 relationships:
Instructor-**Teaches**-Course, Student-**Registers**-Course, Course-**Allocated**-Classroom, Instructor-**Guides**-Student.

- (i): Registration is drawn as an entity and Student as a relationship — wrong.
- (ii): no Student entity; an extra Projects entity is used — wrong.
- (iii): Classroom is attached directly to Course and the Allocation relationship does not connect Classroom — wrong.
- (iv): all four entities and all four relationships are connected correctly.

Answer: (iv)

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Q11. Answer: A

In a B+ tree of order m, every node (internal or leaf) must be at least half full — **except the root**.

The root can have as few as 2 children (or just 1 key if it is a leaf), otherwise a small tree could never be built.

Answer: Only the root node

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Q12. Answer: A, C

1NF only requires every attribute value to be **atomic** (no multi-valued or composite attributes).

- R can have a multi-attribute key — TRUE (e.g. key (roll_no, course_id)). This is about keys, not about values.
- R cannot have a foreign key — FALSE, 1NF puts no restriction on foreign keys.
- R cannot have a composite attribute — TRUE, a composite attribute is non-atomic and must be split into simple attributes.
- R cannot have more than one candidate key — FALSE.

Answer: R can have a multi-attribute key; R cannot have a composite attribute

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Q13. Answer: C, D

Checking each statement:
- $L_1 \cap \overline{L_2} = \phi$ only means $L_1 \subseteq L_2$. Equality also needs $L_2 \subseteq L_1$ — FALSE.
- $L_1 \cup L_3$ can be regular: take $L_1 = \Sigma^*$, then $L_1 \cup L_3 = \Sigma^*$ — FALSE (not always).
- If $\overline{L_3}$ were regular, then its complement L_3 would be regular (closure under complement). So $\overline{L_3}$ is not regular — TRUE.
- Regular languages are closed under complement and union, so $\overline{L_1} \cup \overline{L_2}$ is regular — TRUE.

Answer: $\overline{L_3}$ is not regular; $\overline{L_1} \cup \overline{L_2}$ is regular

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Q14. Answer: D

Checking each statement:
- Threads can also be implemented in user space (user-level threads) — FALSE.
- The file descriptor table belongs to the process and is shared by all its threads — FALSE.
- Every thread has its own stack (and registers, PC) — FALSE.
- Threads share the same address space, so one thread can read/overwrite another thread's data; there is no protection between them — TRUE.

Answer: Threads belonging to a process are by default not protected from each other

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Q15. Answer: B, C

Checking each statement:
- Running → Ready: possible (preemption / time slice over).
- **Waiting → Running: NOT possible.** When the I/O finishes, the process goes to Ready; only the scheduler can move it to Running.
- **Ready → Waiting: NOT possible.** A process can block (request I/O) only while it is running.
- Running → Terminated: possible (exit).

Answer: Waiting to Running, Ready to Waiting

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Q16. Answer: A, D

Bottom-up parsers build the parse tree from the leaves up to the root using **reductions**: shift-reduce parsers, operator precedence parser and the LR family (LR(0), SLR, LALR, CLR).

Predictive parser and LL(1) parser are **top-down** parsers.

Answer: Shift-reduce Parser, LR Parser

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Q17. Answer: B, C

Checking each statement:
- Independence: $P(A)P(B) = 0.15 \neq 0.1$ — FALSE.
- $P(A \cup B) = 0.3 + 0.5 - 0.1 = 0.7$ — TRUE.
- $P(A \cap B^c) = P(A) - P(A \cap B) = 0.3 - 0.1 = 0.2$ — TRUE.
- $P(A^c \cap B^c) = 1 - P(A \cup B) = 0.3 \neq 0.4$ — FALSE.

Answer: $P(A \cup B) = 0.7$ **and** $P(A \cap B^c) = 0.2$

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Q18. Answer: A, C

The circuit computes $Y = X \cdot \overline{X}$.

- No delays: $X \cdot \overline{X} = 0$ always $\rightarrow$ Y is always 0 (option 1 TRUE, option 2 FALSE).
- X goes 0 $\rightarrow$ 1: the AND gate sees X = 1 immediately, but the NOT output is still 1 until its delay passes. For that short time both inputs are 1, so Y gives a **transient 1 (glitch)** — TRUE.
- X goes 1 $\rightarrow$ 0: the direct input becomes 0 immediately, so Y stays 0 no matter what the NOT output does — no transient, FALSE.

Answer: Y is always 0 with no delays; with delays Y can have a transient 1 after X goes 0 $\rightarrow$ 1

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Q19. Answer: A, D

Checking each statement:
- The initial sequence number (ISN) is chosen randomly (to avoid confusion with old segments and for security) — TRUE; it is not always 0.
- A SYN segment consumes one sequence number, so Q acknowledges with the next expected number: $N_Q = N_P + 1$ — TRUE (so $N_Q = N_P$ is false).

Answer: N_P **is chosen randomly by P;** $N_Q = N_P + 1$

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Q20. Answer: A, B, C

Checking each statement:
- Option 1: this is exactly the definition of forwarding (bypassing) — TRUE.
- Option 2: a value available at the output of MEM (e.g. a loaded value) can be fed to the EX input of a later instruction — TRUE.
- Option 3: forwarding cannot remove every stall. Example: *load* followed immediately by an instruction using the loaded value still needs 1 stall (load-use hazard) — TRUE.
- Option 4: forwarding needs extra hardware (extra paths, multiplexers, hazard detection logic) — FALSE.

Answer: Options 1, 2 and 3

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Q21. Answer: A, C

For an outgoing packet, NAT replaces the private **source IP** (and port) with its public IP.

Because a header field changed, the **header checksum** must be recomputed.

Destination IP is unchanged (it is the outside server), and total length is unchanged (payload size is the same).

Answer: Source IP, Header Checksum

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Q22. Answer: 2

Bijection $A \to B \Rightarrow |A| = |B| = n$.
Bijection $A \times A \to A \cup B \Rightarrow n^2 = |A \cup B|$.

But $n \leq |A \cup B| \leq 2n$, so $n \leq n^2 \leq 2n \Rightarrow n \leq 2$.

- $n = 1$: $A = B$, $|A \cup B| = 1 = 1^2$ True
- $n = 2$: take A and B disjoint, $|A \cup B| = 4 = 2^2$ True

So $|A| \in \{1, 2\}$.

Answer: 2

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Q23. Answer: 6

Precedence: + (highest) $>$ $-$ $>$ * $>$ / (lowest). The lowest-precedence operator is applied last, so / splits the expression:
$(3 + 1 + 5 * 2) \ / \ (7 + 2 - 4 - 7 - 6) \ / \ 2$

First part: + binds before *, so $(3 + 1 + 5) * 2 = 9 * 2 = 18$.
Second part: + first: $7 + 2 = 9$. Then $-$ is right associative: $9 - (4 - (7 - 6)) = 9 - (4 - 1) = 6$.

/ is right associative: $18/(6/2) = 18/3 = 6$.

Answer: 6

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Q24. Answer: 16

Number of spanning trees of a complete graph K_n (Cayley's formula) $= n^{n-2}$.

$4^{4-2} = 16$

Answer: 16

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Q25. Answer: 2

Join on $R.A = S.A$:
- A = 10: (10, 20, 90) $\rightarrow 20 < 90$ True
- A = 30: (30, 40, 45) $\rightarrow 40 < 45$ True
- A = 30: (30, 50, 45) $\rightarrow 50 < 45$ False

(A = 20, 50 in R and A = 40 in S have no match.)

Answer: 2

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Q26. Answer: B

Number of chunks $= 10^6/10^3 = 1000$. Each chunk $= 8000$ bits, so transmission time on a link $= \frac{8000}{10^6} = 8$ ms.

P sends the chunks back to back: the last chunk is fully sent by P at $1000 \times 8 \text{ ms} = 8$ s.
Q forwards each chunk while receiving the next, so only the last chunk adds one more transmission of 8 ms on Q-R.

Total $= 8 + 0.008 = 8.008$ s.

Answer: 8.008

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Q27. Answer: A

Input "MMLK" $\Rightarrow$ D = "MM", H = "L", T = ϵ, U = "K".

- D: $5 + 5 + (-5) = 5$
- H: $5 * 10 + (-10) = 40$
- $T = \epsilon$: -5
- U: 5

$S.val = 5 + 40 - 5 + 5 = 45$

Answer: 45

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Q28. Answer: A

Use the given FIRST and FOLLOW sets:
- $FIRST(S) = \{c, d, f\}$ but daT gives only d, so (1) must give c and f. $S \to Rf$ gives $(FIRST(R) - \epsilon) \cup \{f\} = \{c, f\}$ True ($S \to fR$ gives only f).
- $\epsilon \in FIRST(T)$, so (2) is $T \to \epsilon$.
- $FIRST(R) = \{c, \epsilon\}$, so (3) starts with c.

Check FOLLOW with $R \to cTR$: $FOLLOW(T) \supseteq FIRST(R) - \epsilon = \{c\}$ and $FOLLOW(R) = \{f\}$ (from $S \to Rf$); together with \$ this gives $FOLLOW(S) = FOLLOW(T) = \{c, f, \$\}$ True. With $R \to cR$, c could never follow T.

Answer: (1) $S \to Rf$ **(2)** $T \to \epsilon$ **(3)** $R \to cTR$

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Q29. Answer: D

Leaders: first instruction (L1), targets of jumps (L3, L4, L11) and instructions right after a conditional jump (L10, L13, L15).

Basic blocks:
- B1: L1-L2 (2)
- B2: L3 (1)
- B3: L4-L9 (6)
- B4: L10 (1)
- B5: L11-L12 (2)
- B6: L13-L14 (2)
- B7: L15 (1)

7 basic blocks; the largest (B3) has 6 instructions.

Answer: 7 and 6

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Q30. Answer: A

Final a depends only on which a-update runs first; final b only on which b-update runs first.
- a: $(1+1)\times 2 = 4$ if T1's update is first, $2\times 1+1 = 3$ otherwise.
- b: $(1+1)\times 2 = 4$ if T1's update is first, 3 otherwise.

- T1 then T2: (4, 4). T2 then T1: (3, 3).
- T1: a=a+1, T2: b=2b, T2: a=2a, T1: b=b+1 $\rightarrow$ (4, 3).
- (3, 4) is impossible: it needs T2's a-update before T1's a-update and T1's b-update before T2's b-update. But in program order T1 does a before b and T2 does b before a, so this creates a cycle.

Answer: (a=4, b=4); (a=3, b=3); (a=4, b=3)

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Q31. Answer: D

Build a max-heap bottom-up (heapify from index $\lfloor 10/2 \rfloor = 5$ down to 1):
[82, 101, 90, 11, 111, 75, 33, 131, 44, 93]

- i=5 (111): child 93 $\rightarrow$ no change.
- i=4 (11): children 131, 44 $\rightarrow$ swap with 131.
- i=3 (90): children 75, 33 $\rightarrow$ no change.
- i=2 (101): children 131, 111 $\rightarrow$ swap with 131 (101 then fine above 11, 44).
- i=1 (82): swap with 131, then with 111, then with 93.

Heap: [131, 111, 90, 101, 93, 75, 33, 11, 44, 82]

Answer: 131, 111, 90

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Q32. Answer: A

Divide both sides by n: $\frac{T(n)}{n} = \frac{T(\sqrt{n})}{\sqrt{n}} + 1$.

Let $S(n) = \frac{T(n)}{n}$. Then $S(n) = S(\sqrt{n}) + 1$.

Taking square roots repeatedly, $n \rightarrow n^{1/2} \rightarrow n^{1/4} \rightarrow \cdots$ reaches a constant after $\log \log n$ steps, adding 1 each step. So $S(n) = \Theta(\log \log n)$.

$T(n) = n\, S(n) = \Theta(n \log \log n)$

Answer: $T(n) = \Theta(n \log \log n)$

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Q33. Answer: A

In a min-heap every parent is smaller than its children, so the maximum element cannot have a child — it must be a **leaf**. Any leaf can hold it.

In an array heap of n elements, leaves are at indices $\lfloor n/2 \rfloor + 1$ to n.

$n = 105$: leaves are 53 to 105 $\rightarrow 105 - 52 = 53$ positions.

Answer: 53

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Q34. Answer: B, C, D

Checking each statement:
- $(X, Y) \rightarrow (Z, W)$ does not give $X \rightarrow (Z, W)$; Y may be required — FALSE.
- Decomposition rule: $XY \rightarrow ZW \Rightarrow XY \rightarrow Z$ — TRUE.
- Pseudo-transitivity: $W \rightarrow Y \Rightarrow XW \rightarrow XY$ (augment by X), and $XY \rightarrow Z$, so $XW \rightarrow Z$ — TRUE.
- Transitivity: $X \rightarrow Y,\ Y \rightarrow Z \Rightarrow X \rightarrow Z$ — TRUE.

Answer: Options 2, 3 and 4

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Q35. Answer: C

In a BFS tree, every edge $u \rightarrow w$ of G satisfies $level(w) \leq level(u) + 1$.

A **forward edge** goes from u to a non-child descendant w, so $level(w) \geq level(u) + 2$ — this contradicts BFS. Hence there are no forward edges — TRUE.

The others can fail:
- Back edge: G = {v→a, a→v}. Tree {v→a} is both DFS and BFS tree; a→v is a back edge.
- Cross edge: G = {v→a, v→b, b→a}. DFS visits a then b; b→a is a cross edge, and the tree is also the BFS tree.
- So G can also have edges outside T.

Answer: There are no forward-edges in G with respect to T

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Q36. Answer: B, C, D

Conflicting pairs (same item, different transactions, at least one write):
- z: only T1 uses it → T1 has no conflict with anyone.
- x: only T2 uses it.
- y: $r_3(y), w_3(y)$ come before $r_2(y), w_2(y)$ → **T3 must come before T2**.

So any serial order with T3 before T2 is conflict equivalent, and T1 can be anywhere:
$T_1T_3T_2,\ T_3T_2T_1,\ T_3T_1T_2$.
$T_1T_2T_3$ puts T2 before T3 → not equivalent.

Answer: $T_1T_3T_2$**,** $T_3T_2T_1$**,** $T_3T_1T_2$

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Q37. Answer: A, B

Minterms 3, 5, 6, 7 = 011, 101, 110, 111 $\rightarrow$ F = 1 when at least two inputs are 1 (majority function).

- Maxterms are the remaining combinations: $\Pi(0, 1, 2, 4)$ — TRUE.
- Majority function $= XY + YZ + XZ$ — TRUE.
- Depends on Y: $F(1, 0, 0) = 0$ but $F(1, 1, 0) = 1$ — so not independent of Y.
- Depends on X: $F(0, 1, 0) = 0$ but $F(1, 1, 0) = 1$ — so not independent of X.

Answer: $F = \Pi(0, 1, 2, 4)$ **and** $F = XY + YZ + XZ$

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Q38. Answer: B, D

Each iteration does $x \leftarrow 2x + y$, repeated y times. Solving:
$f(x, y) = 2^y x + (2^y - 1)y = 2^y(x + y) - y$

- $x = 20, y = 10$: $1024 \times 30 - 10 = 30710$. This is less than 2^{20} but more than 2^{10} $\rightarrow$ options 1 and 3 FALSE.
- $x = 20, y = 20$: $2^{20} \times 40 - 20 > 2^{20}$ $\rightarrow$ TRUE.
- $x = 10, y = 20$: $2^{20} \times 30 - 20 > 2^{20}$ $\rightarrow$ TRUE.

(Both values are about 4×10^7, so no int overflow.)

Answer: Options 2 and 4

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Q39. Answer: A

$rank(A) \leq n < m$, so the solution space has dimension $m - rank(A) \geq m - n$.

- At least $m - n$ linearly independent solutions — TRUE.
- "$m - n$ vectors span every solution" fails when the dimension is larger, e.g. A = zero matrix (dimension m) — FALSE.
- $A = [1 \;\; 1]$ ($n = 1, m = 2$): solutions are $t(1, -1)$; no non-zero solution has a zero entry, though $m - n = 1$ — FALSE.
- $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$: solutions $(0, 0, t)$ have only 1 non-zero variable $< n = 2$ — FALSE.

Answer: There exist at least $m - n$ linearly independent solutions

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Q40. Answer: B, C

Reading 0 moves 1→2→3→1, reading 1 moves 1→4→5→1, and the reverse arrows undo a step. So the DFA tracks $(n_0 - n_1) \bmod 3$:
state 1 = 0, states 2 and 5 = 1, states 3 and 4 = 2.

$L(M) = \{w : n_0(w) \equiv n_1(w) \pmod 3\}$, and states with the same value are equivalent.

- 2 and 4 (values 1 and 2): distinguishable (input "1": 2→1 accept, 4→5 reject) — TRUE.
- **3 and 4**: both value 2 → equivalent — statement FALSE.
- **2 and 5**: both value 1 → equivalent — statement FALSE.
- $n_0 = n_1 \Rightarrow$ difference 0 $\Rightarrow$ accepted — TRUE.

Answer: "States 3 and 4 are distinguishable" and "States 2 and 5 are distinguishable"

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Q41. Answer: B, C

Checking each statement:
- Complete subgraph of size k: FALSE. The 5-cycle C_5 has $\chi = 3$ but no triangle.
- Independent set of size $\geq n/k$: TRUE. The k colour classes are independent sets covering n vertices, so the largest has at least n/k vertices.
- At least $k(k-1)/2$ edges: TRUE. In a minimum colouring, every pair of colour classes must have an edge between them (otherwise merge the two classes and use $k-1$ colours). That gives $\binom{k}{2}$ edges.
- Vertex of degree $\geq k$: FALSE. K_k has $\chi = k$ but every degree is $k-1$.

Answer: Options 2 and 3

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Q42. Answer: B, D

Checking each statement:
- $\diamond$ associative? $(a \diamond b) \diamond c = a + 2b + 2c$, $a \diamond (b \diamond c) = a + 2b + 4c \rightarrow$ NO.
- $\square$ is ordinary multiplication → associative, YES.
- $\diamond$ over $\square$: $a \diamond (b \square c) = a + 2bc$, but $(a \diamond b) \square (a \diamond c) = (a+2b)(a+2c)$. For $a = b = c = 1$: 3 vs 9 → NO.
- $\square$ over $\diamond$: $a \square (b \diamond c) = a(b + 2c) = ab + 2ac = (a \square b) \diamond (a \square c) \rightarrow$ YES.

Answer: $\square$ is associative; $\square$ distributes over $\diamond$

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Q43. Answer: B, C

Checking each statement:
- WBC read miss: the LRU victim may be dirty and then it must be written back — FALSE ("never" is wrong).
- WTC read miss: with write-through, memory is always up to date, so a victim never needs to be written back — TRUE.
- WBC write hit: the block becomes dirty, so the dirty bit can change from 0 to 1 — TRUE.
- WTC write miss: the victim block is never dirty, so it is never written back — FALSE.

Answer: Options 2 and 3

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Q44. Answer: 4096

Capacity = cylinders $\times$ surfaces $\times$ sectors per track $\times$ bytes per sector.

$512\ \text{GB} = 2^{39}$, 32 surfaces $= 2^5$, 4096 sectors $= 2^{12}$, 1024 bytes $= 2^{10}$.

$$\text{Cylinders} = \frac{2^{39}}{2^5 \times 2^{12} \times 2^{10}} = 2^{12} = 4096$$

Answer: 4096

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Q45. Answer: 3

Serial part $= 10\%$ of 100 ns = 10 ns, parallel part = 90 ns.
With p cores: $T(p) = 10 + \frac{90}{p} + 10(p-1)$

- $p = 1$: 100 ns
- $p = 2$: $10 + 45 + 10 = 65$ ns
- $p = 3$: $10 + 30 + 20 = 60$ ns
- $p = 4$: $10 + 22.5 + 30 = 62.5$ ns

Minimum at $p = 3$ (also from $\frac{90}{p} = 10p \Rightarrow p = 3$).

Answer: 3

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Q46. Answer: 3

Every instruction is fetched (I-cache) and 25% also access data (D-cache).

Stall cycles per instruction:
- I-cache: $1 \times 0.02 \times 100 = 2$
- D-cache: $0.25 \times 0.08 \times 100 = 2$

Actual CPI $= 2 + 2 + 2 = 6$, perfect-cache CPI $= 2$.

Speedup $= 6/2 = 3.00$

Answer: 3

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Q47. Answer: 14

In each iteration **every** existing process calls fork(), so the number of processes doubles, and each of them prints once. (wait() only changes the order, not the count; each child continues the loop with the current x.)

- x = 3: 1 $\rightarrow$ 2 processes, prints 2
- x = 2: 2 $\rightarrow$ 4 processes, prints 4
- x = 1: 4 $\rightarrow$ 8 processes, prints 8

Total $= 2 + 4 + 8 = 14$

Answer: 14

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Q48. Answer: 40

Use longest prefix match. Address ranges (last byte):
/24 $\rightarrow$ 0-255 (R1), .128/25 $\rightarrow$ 128-255 (R2), .64/26 $\rightarrow$ 64-127 (R3), .192/26 $\rightarrow$ 192-255 (R4).

- 10.1.1.16 $\rightarrow$ only /24 $\rightarrow$ R1
- 10.1.1.72 $\rightarrow$ /26 (64-127) $\rightarrow$ R3
- 10.1.1.132 $\rightarrow$ /25 $\rightarrow$ **R2**
- 10.1.1.191 $\rightarrow$ /25 (not in 192-255) $\rightarrow$ **R2**
- 10.1.1.205 $\rightarrow$ /26 (192-255) $\rightarrow$ R4

R2 gets 2 hosts $\times$ 20 packets $= 40$.

Answer: 40

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Q49. Answer: 179

CNF rules are $A \to BC$ or $A \to a$.

- Each $A \to BC$ increases the number of symbols by 1. To grow from 1 symbol (S) to n symbols needs $n-1$ such steps.
- Each of the n variables is then turned into a terminal by $A \to a$: n steps.

Total $= 2n-1$. Here $n = |w| = 90$: $2(90)-1 = 179$.

Answer: 179

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Q50. Answer: 60

DFS produces one tree per connected component. A tree with k vertices has $k-1$ edges, so a forest with n vertices and c trees has $n-c$ edges.

$100 - c = 40 \Rightarrow c = 60$

Answer: 60

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Q51. Answer: 44

Total strings of length 0 to 5: $1+2+4+8+16+32 = 63$.

- $r = 0^* + 1^*$: ϵ, plus all-0 and all-1 strings of each length 1-5: $1 + 2 \times 5 = 11$.
- $s = 01^* + 10^*$: for each length 1-5, the strings $01\ldots1$ and $10\ldots0$. Length 1 ("0", "1") is already in r, so new ones = $2 \times 4 = 8$.

In r or s: $11 + 8 = 19$. Neither: $63 - 19 = 44$.

Answer: 44

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Q52. Answer: 6596

Page size $= 2\ \text{KB} = 2048$.

$2500 = 1 \times 2048 + 452 \rightarrow$ page 1, offset 452.
Page 1 is in frame 3.

Physical address $= 3 \times 2048 + 452 = 6144 + 452 = 6596$

Answer: 6596

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Q53. Answer: 0.37 to 0.38

Given the first ball is red, 9 red and 15 blue (24 balls) remain. Both are red iff the second is red:

$$P = \frac{P(\text{both red})}{P(\text{first red})} = \frac{\frac{10}{25} \cdot \frac{9}{24}}{\frac{10}{25}} = \frac{9}{24} = 0.375$$

Answer: 0.375

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Q54. Answer: 4

Look at each multiplexer separately:
- M1: both inputs $X1 = X2 = 1$, so $Q1 = 1$ whatever A is.
- M2: both inputs $X3 = X4 = 0$, so $Q2 = 0$ whatever B is.
- M3: $Y = Q1 = 1$ when $C = 0$, and $Y = Q2 = 0$ when $C = 1$.

So $Y = 1$ exactly when $C = 0$; A and B are free: $2 \times 2 = 4$ combinations.

Answer: 4

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Q55. Answer: 6

Data in datagram $= 1420 - 20 = 1400$ bytes. Fragment data (except the last) must be a multiple of 8.

Link 1 (MTU 542): max data $= 522 \rightarrow$ use 520. Fragments: 520, 520, 360.
Link 2 (MTU 360): max data $= 340 \rightarrow$ use 336.
- 520 $\rightarrow$ 336 + 184
- 520 $\rightarrow$ 336 + 184
- 360 $\rightarrow$ 336 + 24

Fragments are reassembled only at the receiver, so it receives $2 + 2 + 2 = 6$ fragments.

Answer: 6

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Q56. Answer: A

dry $\rightarrow$ arid $\rightarrow$ parched: increasing degree of lack of water.
diet $\rightarrow$ fast $\rightarrow$ ___ : increasing degree of not eating. Next stronger level is **starve**.

(reject, feast, deny do not fit the scale.)

Answer: starve

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Q57. Answer: D

$x + y = k(x - y)$ for a constant k.

$$x + y = kx - ky \Rightarrow y(1 + k) = x(k - 1) \Rightarrow \frac{x}{y} = \frac{k + 1}{k - 1}$$

This depends only on k, so it is a constant.

Answer: is a constant

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Q58. Answer: A

Sorted: 9, 10, 11, 11, 13, 14, 15, 17, 18, 69 (10 numbers).

Median = average of the 5th and 6th values $= \frac{13 + 14}{2} = 13.5$

Answer: 13.5

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Q59. Answer: C

Let coins be $5k, 3k, 13k$.
Money: Rs.1 $\rightarrow 5k$, Rs.5 $\rightarrow 15k$, Rs.10 $\rightarrow 130k$. Total $= 150k$.

Share of Rs.5 coins $= \dfrac{15k}{150k} \times 100 = 10\%$

Answer: 10%

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Q60. Answer: A

$\log(p^2 + q^2) = \log(9pq) \Rightarrow p^2 + q^2 = 9pq$

Squaring: $p^4 + q^4 + 2p^2q^2 = 81p^2q^2 \Rightarrow p^4 + q^4 = 79p^2q^2$

$$\frac{p^4 + q^4}{p^2q^2} = 79$$

Answer: 79

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Q61. Answer: C

Meaning of each phrase:
- keep his head **down** — stay low-profile / focused
- heading **out** to bat — going out to the pitch
- a head **for** batting — natural talent for it
- a cool head **on** his shoulders — sensible and calm

Answer: (i) down (ii) out (iii) for (iv) on

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Q62. Answer: A

Joining the longer (54 cm) edges gives a tube of height 54 and circumference 4.
$2\pi r = 4 \Rightarrow r = \frac{2}{\pi}, V_{cyl} = \pi r^2 h = \pi \cdot \frac{4}{\pi^2} \cdot 54 = \frac{216}{\pi}$

Cube: $6a^2 = 54 \times 4 = 216 \Rightarrow a = 6, V_{cube} = 216$

Ratio $= \frac{216/\pi}{216} = \frac{1}{\pi}$

Answer: $1/\pi$

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Q63. Answer: C

From the pie chart, fats = unsaturated 20% + saturated 20% + trans 5% = 45%.

Energy from fat $= 0.45 \times 2000 = 900$ kcal.
All fats give 9 kcal/g, so fat $= 900/9 = 100$ g.

Answer: 100

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Q64. Answer: A

Each fold halves the current **longer** side.

- 20 × 8 → fold → 10 × 8
- 10 × 8 → fold → 5 × 8 (now 8 is the longer side)
- 5 × 8 → fold → 5 × 4

Perimeter $= 2(5 + 4) = 18$ cm.

Answer: 18

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Q65. Answer: A

For AB to be a line of symmetry, every square must have its mirror image on the other side of AB.

- Left figure: 1 square above AB and 2 squares below (one just below the line on the left, one two rows down on the right). None of them has its mirror image $\rightarrow$ add 3.
- Right figure: 3 squares, all below AB $\rightarrow$ add their 3 mirror images above.

Total $= 3 + 3 = 6$

Answer: 6

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